Obed Junias

obed.junias@colorado.edu | obedjunias.com | obedjunias19 | obed-junias | 720-266-0046

Research Statement

I study how large language models reason over information and why fluent generation can mask brittle or spurious inference. My work focuses on designing structured evaluation frameworks that expose hidden reasoning failures in multi-step commonsense tasks beyond surface-level accuracy.

A central direction of my research is understanding how training-data distributions shape observed reasoning behavior. In particular, I study how increasing reliance on model-generated data shifts the effective distribution over time, and how structured benchmarks can detect and characterize these shifts.

More broadly, I am interested in how AI systems retrieve, represent, and reason over knowledge, and in building evaluation methods that make these processes observable and reliable for real-world information-seeking systems.

Research & Technical Interests

Research Interests
Commonsense reasoning in LLMs
Robust evaluation of multi-step reasoning
Information retrieval and grounded reasoning
Benchmark design and synthetic reasoning datasets
Human–AI interaction in information-seeking systems

Technical Interests
LLM serving and inference systems
Distributed training and GPU optimization
Multi-agent LLM systems
Retrieval-augmented generation (RAG) systems
LLM evaluation and benchmarking

Education

M.S. in Computer Science, **University of Colorado Boulder** — Aug 2024 – May 2026
Courses: NLP, Deep Natural Language Understanding, Neurosymbolic NLP Methods — (GPA: 4.0/4.0)

B.E. in Computer Science, BMS College of Engineering — Aug 2017 – Aug 2021
Relevant Courses: Algorithms, Databases, Operating Systems, Machine Learning

Publications

- **Junias, Obed***, et al. *Assessing Algorithmic Bias in Language-Based Depression Detection: A Comparison of DNN and LLM Approaches.* IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI), 2025. (Peer-reviewed, Presented). https://ieeexplore.ieee.org/abstract/document/11269509
- **Junias, Obed**, and Maria Leonor Pacheco. *LOGICAL-COMMONSENSEQA: A Benchmark for Logical Commonsense Reasoning.* arXiv preprint arXiv:2601.16504, 2026. https://arxiv.org/abs/2601.16504

Research Experience

Structured Commonsense Reasoning and Benchmarking — June 2025 – Present
Advisor: Maria Leonor Pacheco, BLAST Lab, Associated publication listed above

- Designed **LOGICAL-COMMONSENSEQA**, a benchmark for logical commonsense reasoning, with structurally composed questions and answer options to isolate multi-fact and compositional inference in LLMs.
- Developed evaluation protocols to analyze failure modes of chain-of-thought and few-shot prompting, identifying cases where fluent reasoning masks incorrect logical inference.
- Designing entailment-like reasoning representations to study alignment between predicted answers and underlying reasoning structure.
- Conducting empirical analyses on how reasoning performance varies with task structure and inference depth, forming the basis of an ongoing research manuscript.

Responsible LLM Evaluation in Mental Health Applications — Jan 2025 – Nov 2025
Advisor: Theodora Chaspari, HUBBS Lab, Associated publication listed above

- Built **large-scale LLM evaluation pipelines** to assess demographic bias and subgroup performance of LLMs (e.g., LLaMA, GPT-4) for mental health classification tasks.
- Studied how prompting strategies and fairness-aware losses affect reasoning consistency and robustness in low-resource and imbalanced data settings.
- Developed diagnostic metrics to characterize bias and reasoning instability beyond aggregate accuracy.
- Led the research design and analysis for a first-author, peer-reviewed publication presented at BHI 2025.

Efficient Multi-Agent LLM Inference with Parameter-Efficient Adaptation Aug 2025 – Dec 2025
University of Colorado Boulder

- Designed a modular inference framework to study how parameter-efficient adapters enable scalable multi-agent LLM systems.
- Implemented dynamic **LoRA adapter loading and scheduling** under strict GPU memory constraints, streaming adapters from CPU to GPU at runtime.
- Conducted controlled experiments comparing adapter-based specialization against full-model switching, analyzing latency, throughput, and memory efficiency.
- Demonstrated that adapter-based agents reduce cumulative switching overhead by **20–90**×, enabling complex agent workflows on constrained hardware.

Awards and Recognition

- **NSF–EMBS–Google Young Professional NextGen Scholar**, *IEEE EMBS BHI Conference*, 2025

Relevant Research Projects

Lost in Plot: Contrastive Learning for Logical Semantic Retrieval

- Formulated a semantic retrieval task for "tip-of-the-tongue" queries, targeting underspecified and ambiguous natural-language descriptions requiring semantic rather than lexical matching.
- Designed a BERT-based dual-encoder trained with a multi-task contrastive learning objective to align free-form user descriptions with structured movie metadata.
- Constructed a 100K+ synthetic dataset to enable scalable training and controlled evaluation.
- Evaluated retrieval quality using Recall@K and MRR, outperforming a GPT-4 few-shot baseline.

Medical Ethics Assessment of Large Language Models

- Designed evaluation protocols to assess ethical reasoning of LLMs in medical decision-making scenarios.
- Built a multiple-choice benchmarking pipeline to analyze how external knowledge affects reasoning consistency.
- Analyzed rationales and confidence signals to identify systematic failure modes in ethically sensitive settings.

Industry Experience

Senior Member of Technical Staff at Oracle Corporation Aug 2021 - Aug 2024

- Designed and implemented scalable automation and validation frameworks (**TestNG, BATS**) for end-to-end testing of Oracle REST Data Services (ORDS) and SQLcl, reducing manual validation effort by **95%** and improving correctness guarantees.
- Developed a Python-based migration framework to automate deployment and transfer of Oracle APEX applications, enabling reliable large-scale migrations and supporting 200+ internal users.
- Led technical validation and delivery of a new Oracle Cloud DB Tools feature, coordinating a 15-engineer effort and achieving production readiness within 2 months.

Machine Learning Engineer Intern at Hewlett Packard Enterprise Feb 2021 - Jul 2021

- Built OCR-based automation pipelines for processing unstructured enterprise documents, integrating learned document extraction into production workflows.
- Evaluated model performance across noisy and heterogeneous inputs, contributing to improvements in robustness and downstream automation reliability.

Skills

Languages	Python, SQL, Java, C++
LLM Systems	LLM Serving, LLM Evaluation, RAG Pipelines, Multi-Agent Systems and Tools
LLM Training	Pre-Training, Supervised Fine-Tuning, Transformers, PyTorch DDP
ML Infrastructure	CUDA, FlashAttention, Quantization, GPU Memory Optimization
Frameworks	PyTorch, Triton, Transformers, vLLM, LangChain, LangGraph
Systems & Cloud	Docker, Kubernetes, Redis, Vector Databases, AWS, GCP, OCI